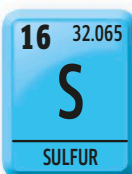
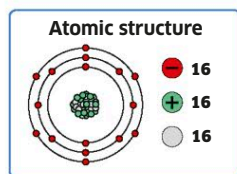


SULFUR

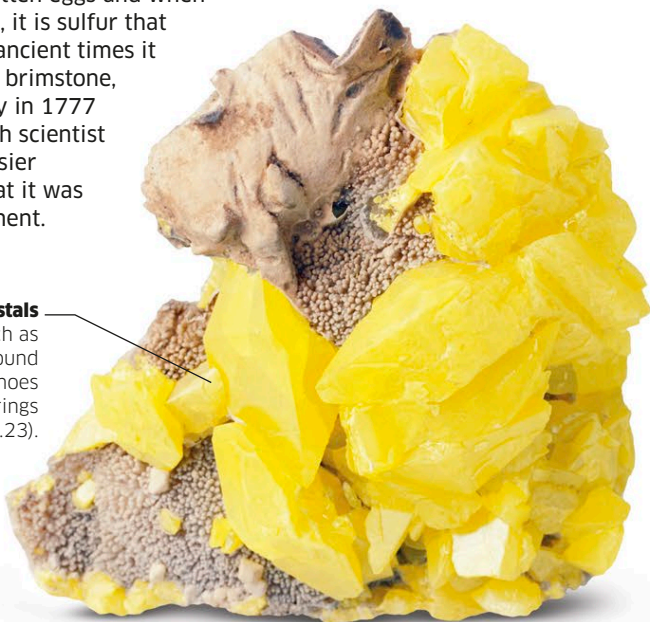
Sulfur

Discovered: 1777

This element has a distinctive yellow color. Many compounds containing sulfur have a strong smell—for example, in rotten eggs and when onions are cut, it is sulfur that is at work. In ancient times it was known as brimstone, but it was only in 1777 that the French scientist Antoine Lavoisier discovered that it was in fact an element.



Sulfur crystals
Crystals such as these can be found near volcanoes and hot springs (see p.23).

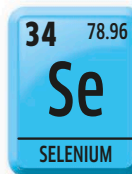
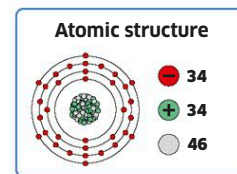


SELENIUM

Selenium

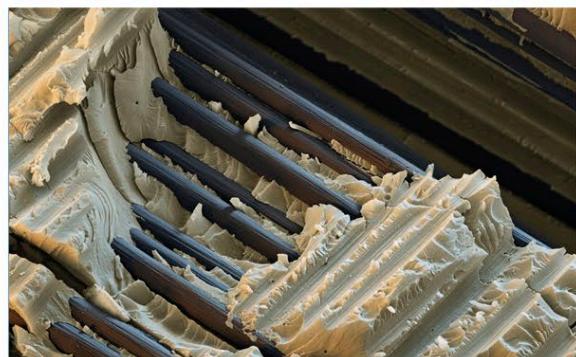
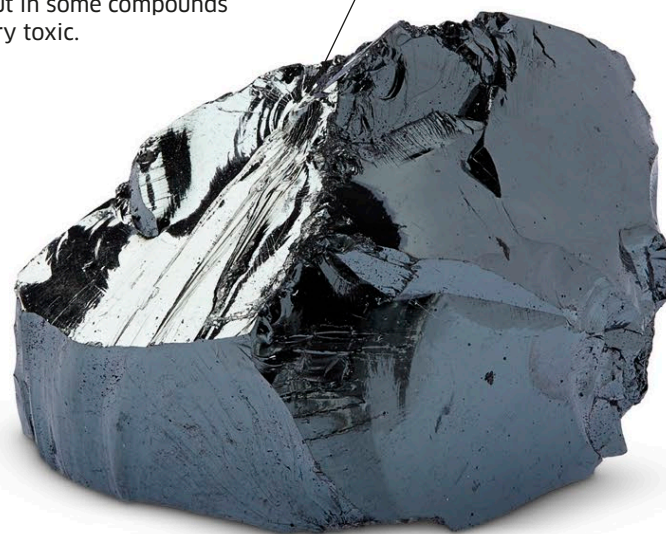
Discovered: 1817

Named after the Greek word *selene*, meaning “moon,” selenium exists in three forms: red, gray, and black selenium. This is an element we need in just the right amount for our bodies to stay healthy, and it is a useful ingredient in anti-dandruff shampoo, but in some compounds it can be very toxic.



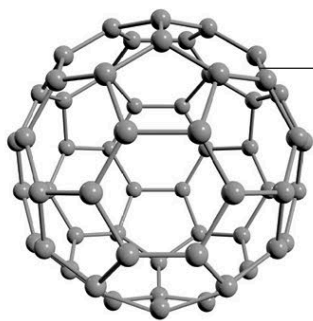
Gray selenium

The most stable form of pure selenium is hard and shiny.



Carbon fiber

In modern materials technology, carbon fibers that are one-tenth of a hair in thickness, but very tough, can be used to reinforce materials such as metals, or plastic (as seen above, enlarged many times).



The carbon atoms are arranged in a rigid, stable structure that looks like a football.

Buckminsterfullerene

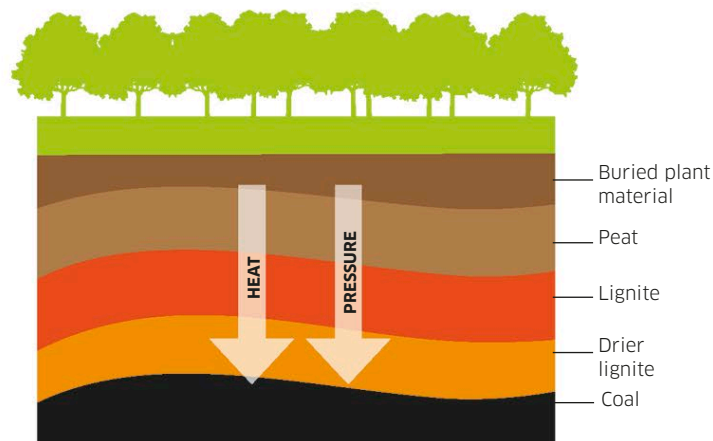
Nicknamed a buckyball, buckminsterfullerene is any spherical molecule of carbon atoms, bonded in hexagons and pentagons. There are typically 60 atoms in a “ball.” They exist in soot, but also in distant stars, and were only discovered in 1985.

Carbon fossil fuels

The substances we call hydrocarbon or fossil fuels include coal, natural gas, and oil. These fuels were formed over millions of years from decaying dead organisms. They are made up mainly of carbon and hydrogen, and when they burn they produce carbon dioxide gas (see p.50–51).

Coal

A long, slow process turned trees that grew on Earth some 300 million years ago into coal that we can mine today. As dead trees fell, they started to sink deep down in boggy soil. They slowly turned into peat, a form of dense soil, which can be burned when dried. Increasing heat and pressure compacted the peat further, turning it into lignite, a soft, brown rock. Even deeper down, the intense heat turned the lignite into solid coal.



Oil and natural gas

The crude oil that is used to make diesel and gasoline is known as petroleum, meaning “oil from the rock.” Millions of years ago, a layer of dead microorganisms covered the seabeds. It was slowly buried under mud and sand, gradually breaking down into hydrocarbons. Heat and pressure changed mud into rock and organic matter into liquid, or gas. This bubbled upward until it reached a “lid” of solid rock, and an oil (or gas) field was formed.

